

ANIMAL DETECTION AND ALERT SYSTEM USING RASPBERRY PI 4

An internship report submitted in partial fulfillment of the requirements for the award
of the degree of

Bachelor of Technology

in

Department Electronics and Communication Engineering

by

NALLAMOTHU BALAKRISHNA (EC23B1033)



**DEPARTMENT OF
ELECTRONICS AND COMMUNICATION ENGINEERING NATIONAL
INSTITUTE OF TECHNOLOGY PUDUCHERRY
KARAIKAL – 609 609**

SEP-NOV 2024

BONAFIDE CERTIFICATE

This is to certify that the internship report entitled "**Animal detection and alert system using Raspberry pi 4** " has been submitted by **Nallamothu Balakrishna**, bearing the register number **EC23B1033**. He has successfully completed the internship at 'National Institute of Technology Puducherry' under the supervision/guidance of **Dr. M Surendar**, Assistant Professor, Department of ECE, affiliated with NITPY, during the period from SEP 2024 to NOV 2024.

The internship report is accepted as a partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in the Department of Electronics and Communication Engineering at the National Institute of Technology, Puducherry, Karaikal, during the academic year 2026-27.

Dr.M Surendar

Internship mentor

Assistant Professor

Dept. of ECE

NITPY, Karaikal

Dr. M Surendar

Internship Coordinator

Assistant Professor

Department of ECE

NITPY, Karaikal

Dr. Malaya Kumar Nath

Head of the Department

Assistant Professor

Department of ECE

NITPY, Karaikal

Internship viva-voce held on _____

ACKNOWLEDGEMENTS

I am grateful to my project supervisor, **Dr Surendar M**, for their unwavering support, insightful feedback, and expert guidance throughout the project. Their expertise and encouragement have been instrumental in the successful completion of this work.

I am grateful to **Dr. Makarand Madhao Ghangrekar**, Director, NIT Puducherry, for providing a conducive environment for learning and research. I extend my heartfelt thanks to **Dr. Sundaravarathan S**, Registrar, NIT Puducherry, for the administrative support throughout the project. I would like to express my gratitude to the Dean, Academics of NIT Puducherry, for the valuable insights, encouragement, and support during this project.

A special word of thanks goes to **Dr. Malaya Kumar Nath**, the Head of the Department of Electronics and Communication for his continuous guidance, mentorship, and encouragement. Thanks are due to our internship coordinator **Dr Surendar M**. His expertise and knowledge have been invaluable in shaping the direction of this internship program.

I would also like to acknowledge the faculty members, mentors, and colleagues who provided valuable suggestions, feedback, and assistance during the course of this project. Their contributions have greatly enriched the project.

Finally, I express my heartfelt appreciation to my friends and family members for their unconditional support, understanding, and motivation throughout this journey. I am truly grateful for the support and encouragement received from each of them, as their contributions have played a vital role in the successful completion of this project.

ABSTRACT

This project develops an automated system for detecting, identifying, and alerting users about wildlife in remote areas using Raspberry Pi, sensors, and machine learning. Traditional wildlife monitoring methods face high costs, limited accessibility, and constant human oversight, which this system overcomes by autonomously detecting animals, capturing images, identifying species, storing data in the cloud, and sending real-time SMS alerts.

The system combines an ultrasonic sensor, a camera module, and a Convolutional Neural Network (CNN) deployed on a Raspberry Pi 4. The ultrasonic sensor detects animal movement, triggering the camera to capture an image. The CNN, trained on 32 animal classes, classifies the image and stores it in the cloud, while sending an SMS alert with the detection's time and date.

The CNN model, trained for 100 epochs, achieved 92% accuracy, ensuring reliable species identification. Testing showed that the system accurately detects animals, captures images, and classifies them with minimal human intervention.

This solution offers a scalable, cost-effective tool for wildlife monitoring, benefiting researchers, safety personnel, and conservationists by enabling efficient, real-time surveillance in hard-to-reach areas.

TABLE OF CONTENTS

CHAPTER .NO	TITLE	PAGE NO.
	ACKNOWLEDGEMENT	3
	ABSTRACT	4
	TABLE OF CONTENTS	5
	LIST OF TABLES	6
1.	INTRODUCTION 1.1 Human-Wildlife conflict 1.2 Project over view 1.3 System functionality 1.4 Project goal	7
2.	METHODOLOGY 2.1 Problem with present system. 2.2 Objectives. 2.3 System overview.	9
3.	REQUIREMENTS 3.1 Hardware requirements. 3.1.1 Raspberry Pi 4 3.1.2 USB web camera 3.1.3 Ultrasonic sensor 3.2 Software requirements. 3.2.1 Operating system 3.2.2 Programming language and IDE 3.2.3 Libraries and Packages 3.2.4 Cloud storage and connectivity tools	12

4.	SETTING UP RASPBERRY PI 4.1 Introduction to Raspberry Pi 4.2 Initial setup of Pi	16
5.	SYSTEM OPERATION FLOW 5.1 Sensor detection phase. 5.2 Image capture and processing. 5.3 Animal detection and identification phase. 5.4 Sending SMS alert. 5.5 Uploading to the cloud (Google Drive)	18
6.	RESULTS	24
7.	ADVANTAGES OF THIS MODEL	25
8.	FUTURE WORKS AND IMPROVEMENTS	26
9.	CONCLUSION	27
10	REFERENCES	28

TABLE OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1.1	Tiger attack on humans	07
2.1	Project module image	09
2.2	Block diagram of model	10
2.3	System flow chart	11
3.1	Raspberry pi model 4	12
3.2	usb web camera	13
3.3	Ultrasonic sensor	13
4.1	Raspberry pi terminal using PUTTY	15
4.2	PUTTY	16
4.3	Remote Desktop window	17
4.4	File sharing	17
5.1	Ultrasonic sensor operation	18
5.2	Ultrasonic sensor with Raspberry pi	19
5.3	USB web cam with raspberry Pi	19
5.4	PI camera interfaced with Pi	20
5.5	Model detecting animal	21
5.6	Predicted animals	22
5.7	Accuracy plot of model	22
5.8	Block diagram of process of sending SMS	23
5.9	Images uploaded in drive	23

CHAPTER 1

INTRODUCTION

Ecology's mission focuses on understanding wildlife in their natural habitats, especially regarding human-induced environmental changes. Rapid population growth and economic development have resulted in resource overexploitation, altering ecosystems and landscapes. As urban areas encroach on forests, wildlife struggles to find food and water, increasing the chances of animals entering human environments.

1.1 Human-Wildlife Conflict

The encroachment into natural habitats has escalated conflicts between humans and wildlife, often resulting in tragic outcomes. In India alone, over the past three years, more than 2,500 individuals have reportedly lost their lives due to animal attacks. This alarming statistic underscores the pressing need for improved wildlife monitoring to safeguard both human lives and biodiversity.

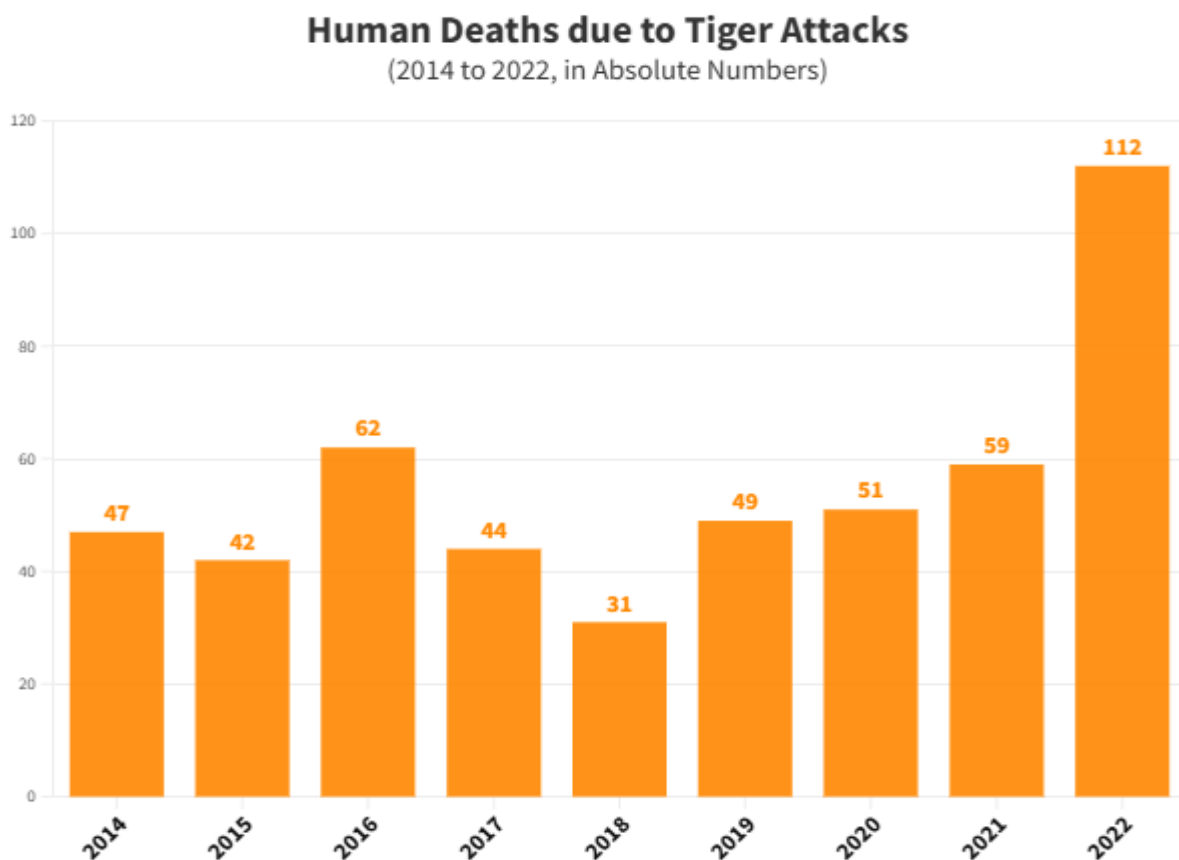


Fig 1.1 Tiger attack on humans

1.2 Project Overview

To tackle these challenges, it is crucial to monitor wildlife in proximity to human settlements. Our project utilizes a Raspberry Pi-based system that features:

- A camera and ultrasonic sensor for detecting animals.
- A machine learning model designed to identify various species, particularly those that may pose threats or are endangered.

1.3 System Functionality

- **Detection and Identification:** When an animal is detected within the sensor's range, the system captures an image and processes it to determine the species.
- **Cloud Storage and Alerts:** The captured image is securely uploaded to the cloud for storage and further analysis. Additionally, an SMS alert is dispatched to local residents and authorities, providing timely notifications about the presence of animals.

1.4 Project Goal

- The primary goal is to monitor and detect wildlife in critical areas and proactively alert residents, reducing the risk of human-animal encounters.
- By using technology to support wildlife conservation and human safety, the project aims to foster a sustainable interaction between humans and nature, maintaining ecological balance amid rapid environmental change.

CHAPTER 2

METHODOLOGY

2.1. Problem with Current Systems:

- Many animals in forests are untracked due to limitations in current monitoring systems.
- Animals that leave forested areas risk danger from human activity, such as crossing roads.
- Food shortages and other environmental challenges negatively affect animals' health, often leading to fatalities.

2.2. Objectives:

- Protect people from potential animal attacks by sending alerts.
- Protect animals from human-caused injuries.
- Protect animals from natural and environmental disasters.

2.3. System Overview:

- A network of cameras is placed at grid points across a forest or nature reserve.
- Images from these cameras are processed by Raspberry Pi modules.



Fig 2.1 Project module image

2.4 Image Processing and Identification:

- Photos are analysed using a Convolutional Neural Network (CNN) model.
- The CNN learns key features from the dataset and identifies animals based on these features.
- Each CNN layer filters results to refine identification.

2.5 Alert System:

- If a potentially dangerous animal is detected, the system sends an alert to the control room via message and email.

2.6 Sensor Integration:

- An Ultrasonic sensor detects the presence of an animal and activates the camera to capture an image

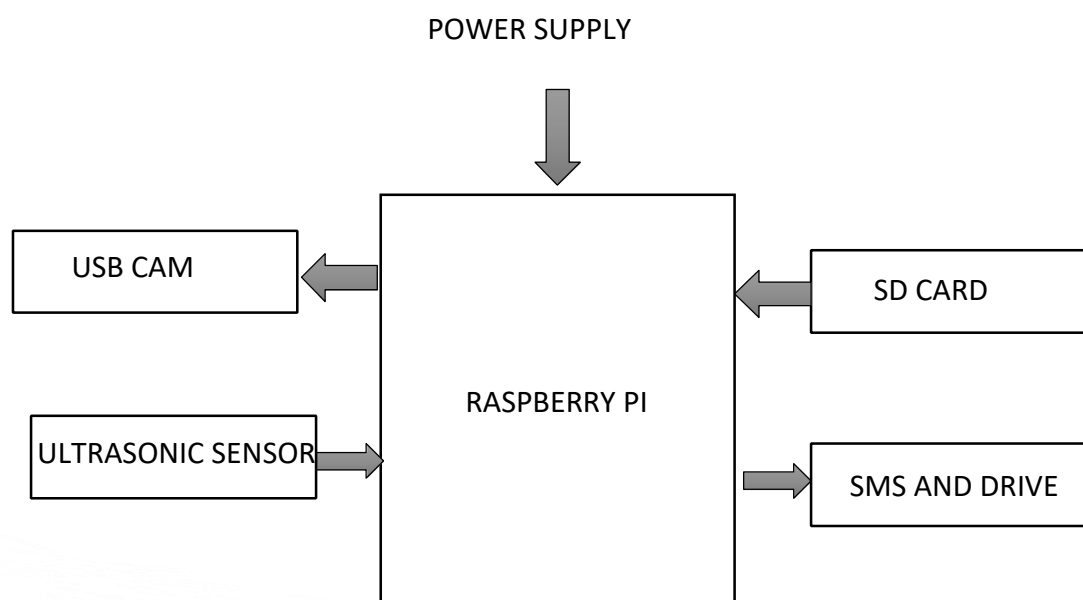


Fig 2.2 Block diagram of model

→In this proposed project, a Raspberry Pi 4 Model B is utilized as the microcontroller to manage the various connected devices. The key hardware components include an ultrasonic sensor to measure distance and detect the presence of wild animals, an ultrasonic sensor for enhanced detection capabilities, and a USB webcam to capture images of the animals.

→Power supply is crucial for the operation of the Raspberry Pi, typically requiring 5V. The USB webcam can be configured to take snapshots or record video based on the specific needs of the project.

Additionally, an SD card is inserted into the Raspberry Pi for data storage, allowing for the capture and retrieval of images.

→Once the ultrasonic or IR sensor detects an animal, it triggers the USB webcam to capture an image. The system is designed to send notifications to the control room via SMS, alerting staff and registered people about the detected animal and ensuring timely responses to potential wildlife interactions.

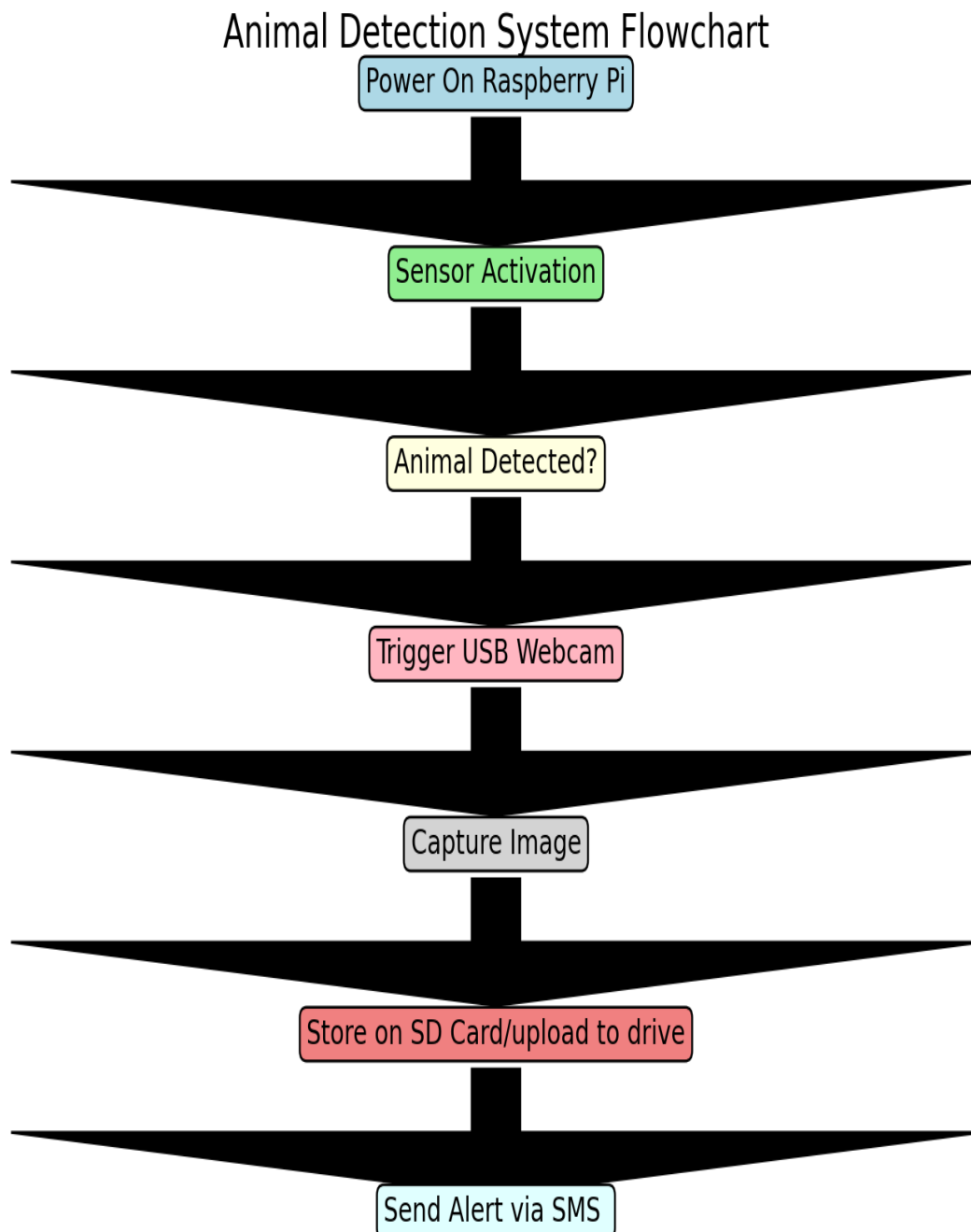


Fig 2.3 System flow chart

CHAPTER 3

REQUIREMENTS

3.1 Hardware requirements:

3.1.1 Raspberry pi 4: The Raspberry Pi is a series of small, affordable computers created by the Raspberry Pi Foundation to promote computer science education and innovation. Over time, it has become widely used in fields like robotics, IoT, and environmental monitoring. Designed to run Linux-based operating systems, Raspberry Pi models provide connectivity options such as USB, HDMI, and Ethernet, along with a 40-pin GPIO (General Purpose Input/Output) interface that connects to external hardware like sensors, cameras, LEDs, and motors.

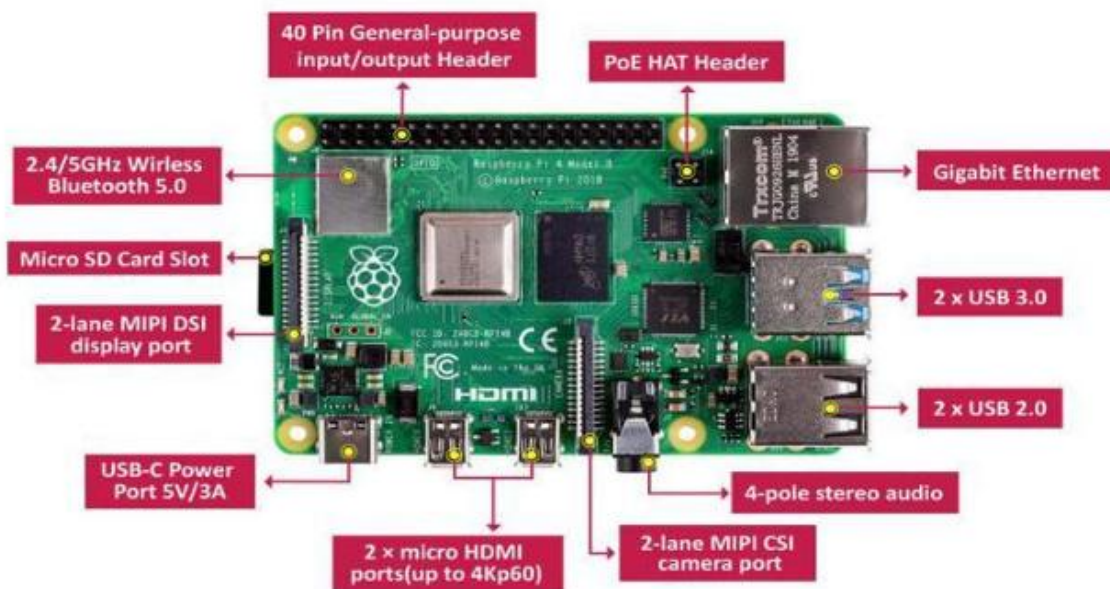


Fig 3.1 Raspberry pi model 4

The Raspberry Pi 4 Model B, one of the latest models, features a 1.5 GHz quad-core ARM Cortex-A72 processor and up to 8 GB of RAM, allowing it to handle complex applications. It supports dual 4K displays via two micro-HDMI ports, has USB 3.0 and USB 2.0 ports for fast data transfer, and provides network options like gigabit Ethernet, Wi-Fi, and Bluetooth 5. Powered through a USB-C port, it is compact, energy-efficient, and capable of powering additional peripherals.

3.1.2 USB Web camera:

A USB webcam is an external camera that connects via USB, providing an easy, plug-and-play solution for capturing video and images. On Raspberry Pi, it offers a versatile alternative to the Pi Camera Module, supporting snapshots and live streaming through tools like fswebcam and motion. This compatibility makes USB webcams ideal for DIY projects like surveillance, video streaming, and animal detection.



Fig 3.2 usb web camera

3.1.3 Ultra sonic sensor:

The Robotbanao Ultrasonic Sensor module includes a transmitter, receiver, and control circuit. It measures distance by emitting high-frequency sound waves and calculating the time it takes for them to bounce back from an object, converting this "time of flight" data into a distance measurement.

The formula to calculate distance using an ultrasonic sensor is:

$$\text{Distance} = (\text{Time} \times \text{speed of sound}) / 2$$

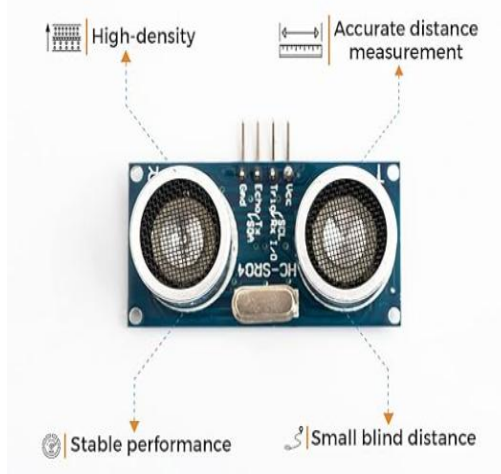


Fig 3.3 Ultrasonic sensor

3.2 Software requirements:

3.2.1 Operating System:

- **Raspberry Pi OS:** A Linux-based OS optimized for Raspberry Pi, providing a user-friendly environment to interact with hardware and run your programs. Install it using the Raspberry Pi Imager tool.

3.2.2 Programming Language:

- **Python:** Primary language for controlling sensors, interfacing with hardware (e.g., ultrasonic sensor), and implementing the machine learning model for animal detection. IDE used for python programming is Thonny

3.2.3 Libraries and Packages:

- **OpenCV:** For image processing and handling webcam input.
- **TensorFlow and Keras:** For running the Convolutional Neural Network (CNN) model to detect and classify animals.
- **Numpy:** For numerical operations in image processing and model handling.
- **Pillow (PIL):** For additional image handling functions, if needed.
- **smtplib (or SMSChef API Integration):** To enable SMS/email notifications in Python.
- **RPi.GPIO:** For controlling GPIO pins and interfacing with the ultrasonic sensor.
- **XRDP** used for remote desktop connection of raspberry pi with laptop or computer.

3.2.4 Cloud Storage and Connectivity Tools

- **Google Drive API:** To store detected animal images in the cloud, ensuring data is securely saved and accessible.
- **SMSChef API:** To send SMS alerts when an animal is detected, helping notify relevant personnel.

CHAPTER 4

SETTING UP RASPBERRY PI

4.1 Introduction to Raspberry pi:

- Raspberry Pi is a versatile single-board computer that supports various applications, from education to IoT. This guide provides step-by-step instructions to set up your Raspberry Pi for remote access and desktop connection using SSH, PuTTY, and VNC.

4.2 Initial set up of Raspberry pi:

1. Installing the operating system:

- Use Raspberry pi Imager to install Raspberry Pi OS on a microSD card.
- Configure Wi-Fi, language, and location during setup.

2. First boot:

- Connect key board, mouse and laptop to configure settings.
- Enable SSH (Secure Shell Protocol) for remote access in Raspberry Pi configuration.

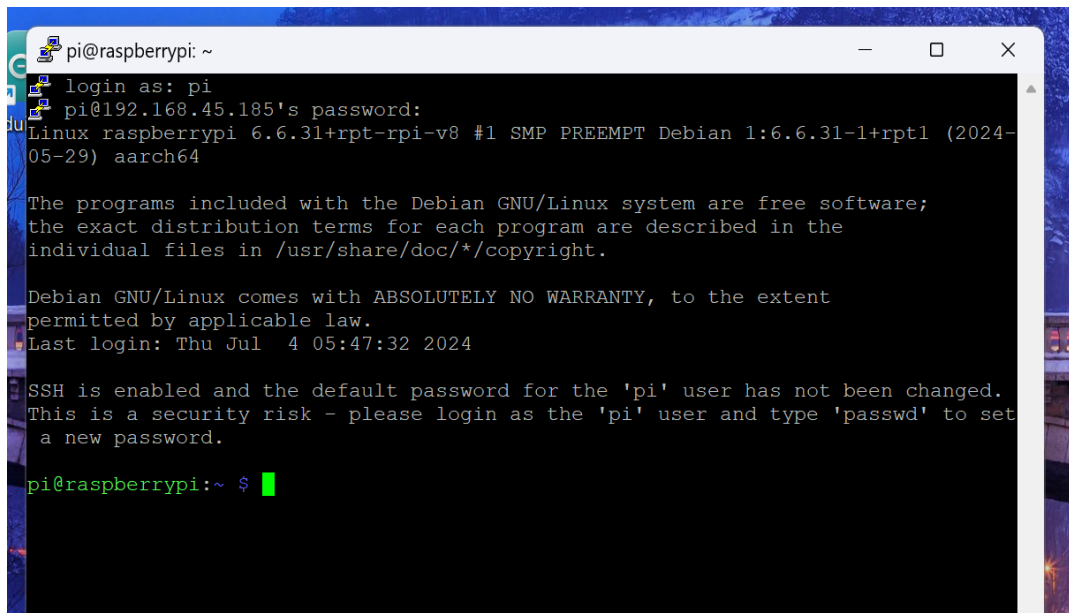


Fig 4.1 Raspberry pi terminal using PUTTY

3.SSH Protocol and Remote access with laptop:

• What is SSH?

- To reach to the Raspberry Pi from PC, we need SSH protocol. SSH is cryptographic network protocol used for remote access of machines in a secure way.

- **Using PUTTY for remote access of Pi terminal:**
 - Download PUTTY, enter the Raspberry Pi's IP address, and connect via SSH.
 - Troubleshoot common issues (e.g. permissions, network).

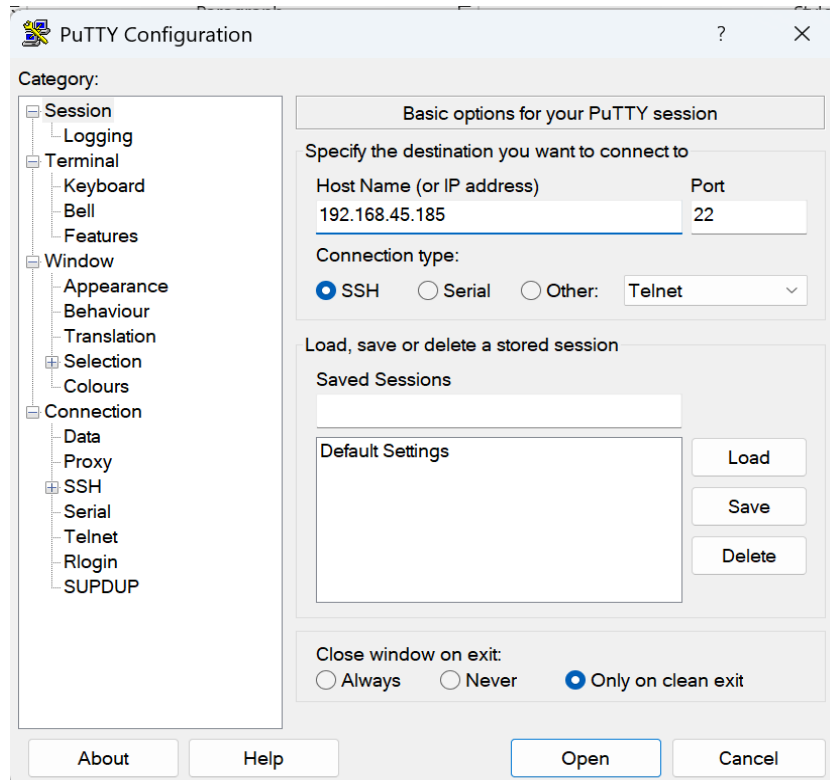
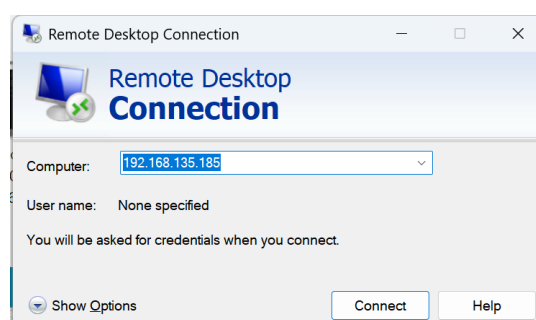


Fig 4.2 PUTTY

4.Setting up Remote Desktop Access:

- Install XRDP (X windows Remote Desktop Protocol) in Raspberry Pi through terminal of raspberry pi.
- Enable XRDP in Raspberry Pi by running the following command in the terminal.
- Sudo install xrdp.
- Sudo systemctl start xrdp
- Now we can go to the Remote Desktop Connection app in our system and enter the IP address of the Raspberry Pi and connect to the Pi.
- A new window of the desktop of Raspberry pi will open.



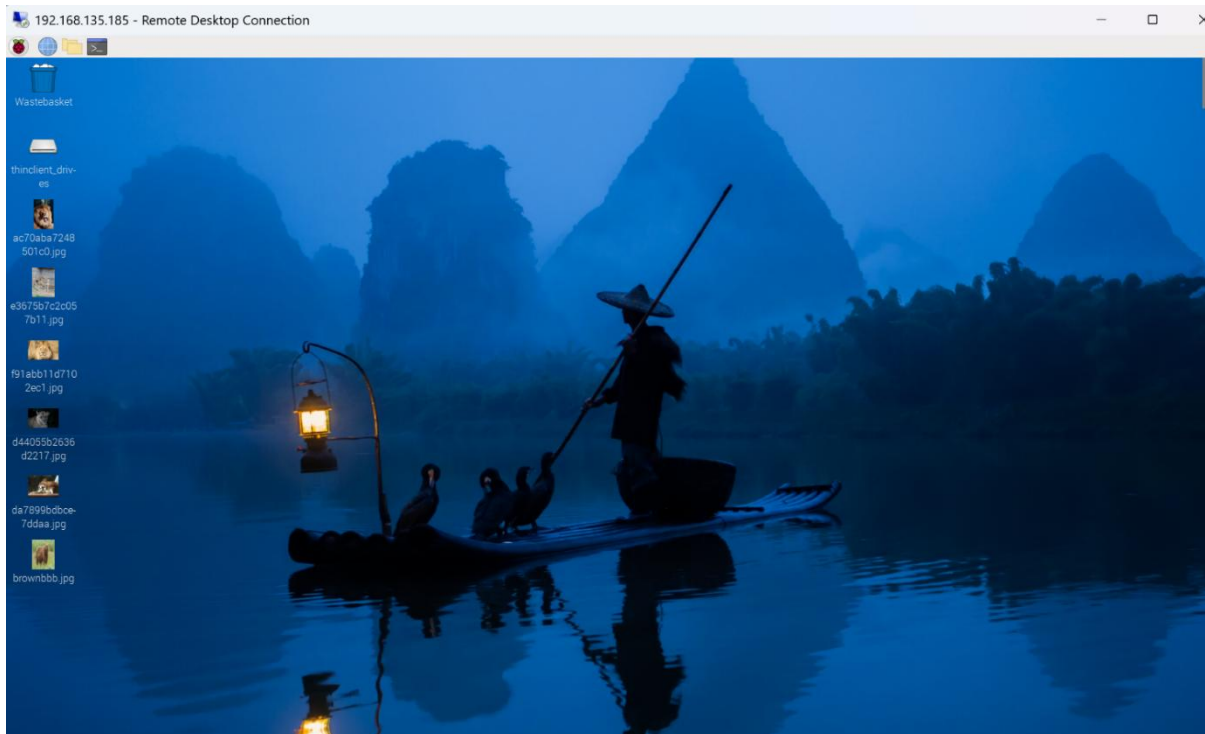


Fig 4.3 Remote Desktop window

5.File transfer between Raspberry Pi and PC:

- **Using SCP (Secure Copy Protocol):**

- Securely transfer files between Raspberry Pi and PC using SCP commands in the terminal/command window.
- Example: 'scp <path of file to be sent> pi@<ip_address>:<path_where_to_be_stored>'

```
C:\Users\balak>scp C:\pictures\lions\ac70aba7248501c0.jpg pi@192.168.135.185:/home/pi/project/
pi@192.168.135.185's password:
ac70aba7248501c0.jpg                                100% 665KB 2.0MB/s 00:00
```

Fig 4.4 File sharing

CHAPTER 5

System Operation Flow

5.1. Sensor Detection Phase

1. **Ultrasonic Wave Emission:** The ultrasonic sensor emits sound waves at a high frequency (inaudible to humans) from its transmitter component.
2. **Echo Reception:** If there's an object within range, the sound waves bounce off and return to the sensor's receiver. The time it takes for this echo to return is measured.
3. **Distance Calculation:** The sensor calculates the distance based on the speed of sound. This data is processed with the formula:

$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2}$$

4. **Range Check:** In our project, a threshold distance is set (like 10 cm). If the change in distance measured is within this range, it indicates an animal might be present.
5. **Trigger Activation:** Once the distance falls within the threshold, the system is triggered. This initiates the camera to capture an image and starts the process of identifying the object using the CNN model.
6. **Continuous Monitoring:** The sensor continues monitoring its surroundings, allowing repeated detection when other objects come into range, ensuring consistent surveillance. This step-by-step process allows precise and real-time monitoring, making it effective for detecting animals and activating the camera only when necessary.



Fig 5.1 Ultrasonic sensor operation

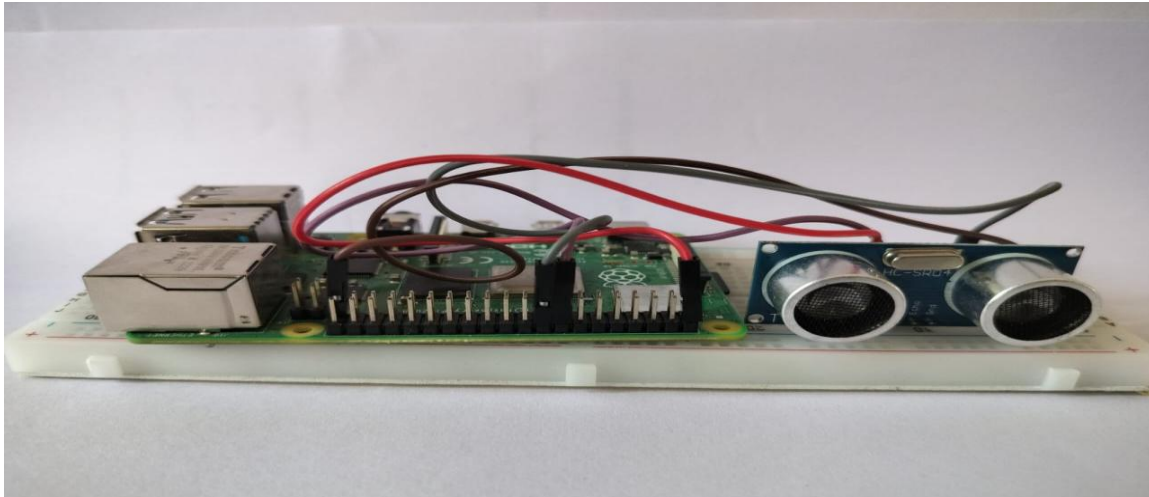


Fig 5.2 Ultrasonic sensor with Raspberry pi

5.2 Image capture and processing

1. Sensor Activation of Camera:

- When the ultrasonic sensor detects an object (animal) within range, it sends a signal to the Raspberry Pi.
- The Raspberry Pi activates the USB webcam to take a snapshot of the detected animal.

2. Image Capture:

- The USB webcam immediately captures an image as triggered.
- This image is saved temporarily on the Raspberry Pi for further processing.



Fig 5.3 USB web cam with raspberry Pi

3.Image Resizing:

- The image is resized to a specific resolution required by the machine learning model (often to smaller dimensions like 224x224 pixels).
- This resizing standardizes the input size for efficient processing.

4. Normalization:

- The pixel values of the image are normalized to a specific range (e.g., 0 to 1) to improve model accuracy.
- Normalization helps the model process images consistently, regardless of lighting or contrast variations.

5.Conversion to Tensor Format:

- The processed image is converted into a tensor format compatible with the CNN model.
- This conversion allows the image data to be directly used as input for the identification model.

6.Ready for Model Input:

- The final pre-processed image is now ready for analysis by the machine learning model, with optimal size, clarity, and normalized values for accurate animal identification.

The Pi Camera is ideal for this project because it offers higher image quality and integrates seamlessly with the Raspberry Pi's software, enhancing the accuracy of animal detection. Its compact design and compatibility with computer vision tasks make it well-suited for fieldwork in remote areas. Additionally, the Pi Camera provides lower latency, ensuring timely capture and processing of images for real-time alerts.



Fig 5.4 PI camera interfaced with Pi

5.3 Animal Detection and Identification Phase

1. Input Image:

- The image is resized to a shape of $224 \times 224 \times 3$ to fit the model's input requirements.

2. Pre-trained VGG19 Model:

- The image passes through the VGG19 layers, which have been trained on ImageNet. These layers extract important features, such as edges, shapes, and textures, from early layers to more complex patterns in deeper layers.

3. Feature Extraction Layer:

- After passing through VGG19's layers (with frozen lower layers), the image's features are flattened and pooled (using GlobalAveragePooling) to reduce dimensions. This step outputs a fixed-length feature vector, representing the high-level features of the image.

4. Custom Layers:

- The flattened vector goes through custom dense layers, which are fully connected layers. Each layer refines the features, learning to recognize the specific animal classes from your dataset.
- Dropout layers prevent overfitting by randomly "dropping" a subset of neurons during training, ensuring the model doesn't rely on specific pathways.

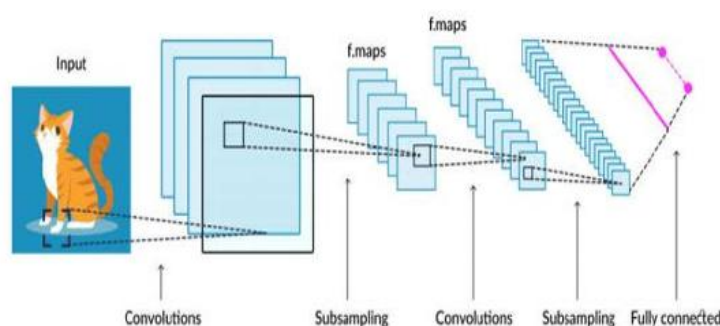


Fig 5.5 Model detecting animal

5. Softmax Layer:

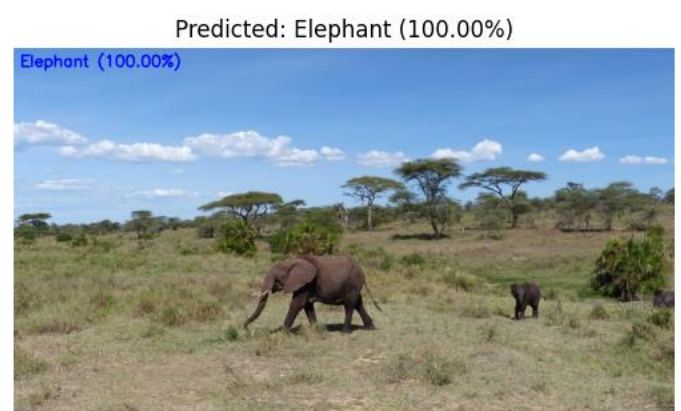
- The final dense layer with a softmax activation outputs a probability distribution across all classes. This layer's output gives the model's confidence levels for each possible class.

6. Prediction:

- The model chooses the class with the highest probability as its predicted label for the input image.



predicted lion



Predicted elephant

Fig 5.6 Predicted animals

The following plot shows the training and validation accuracy of the model that we used.

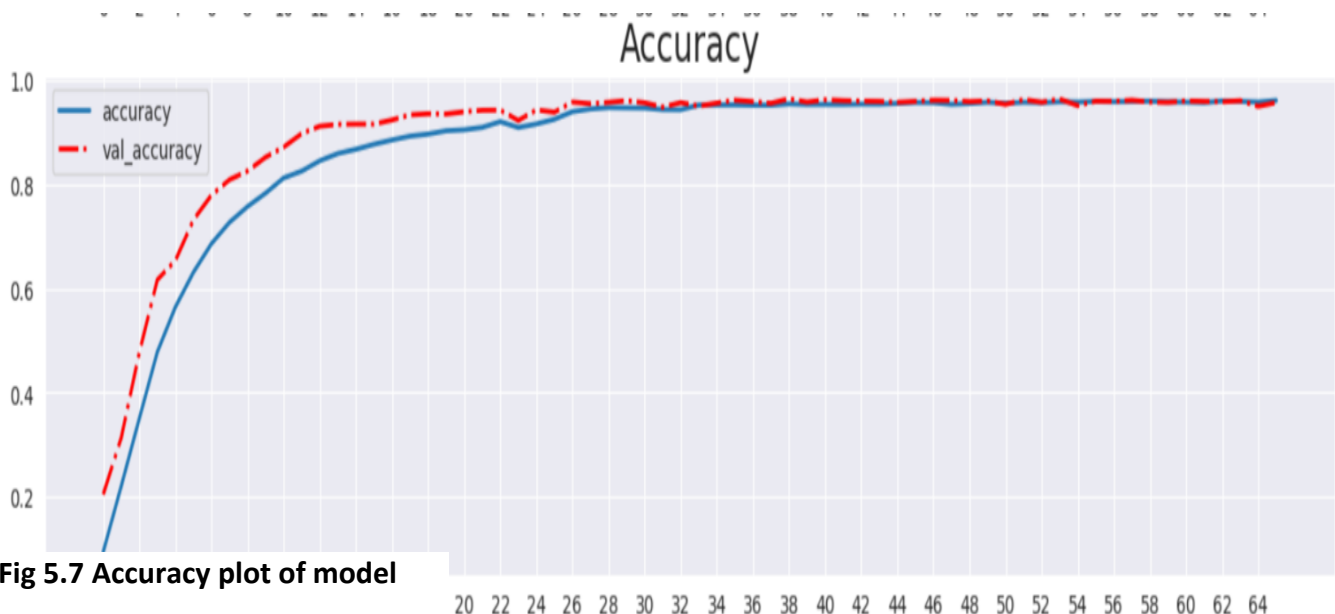


Fig 5.7 Accuracy plot of model

5.4 Sending SMS Alert

- If a recognized animal is detected, the system generates an alert. Using the SMSChef API, the Raspberry Pi sends an SMS to pre-configured phone numbers stored in Google Sheets/list. The SMS alert includes a notification about the detected animal, with the time and date of the detection.
- The API fetches the relevant contacts from the list of phone numbers, formats the alert message, and sends it to notify people immediately.

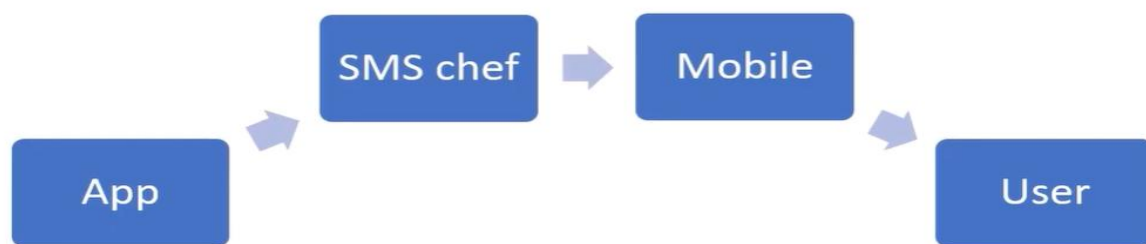


Fig 5.8 Block diagram of process of sending SMS

5.5 Uploading Image to Cloud (Google Drive)

- To ensure the image is stored and accessible for later review, the Raspberry Pi uploads the image to Google Drive. This step uses the **Google Drive API** to create a secure and organized storage location.
- When an image is ready to upload, the Raspberry Pi uses authentication to access Google Drive and save the image in a designated folder. The uploaded image file is timestamped to help with tracking and organizing the data.

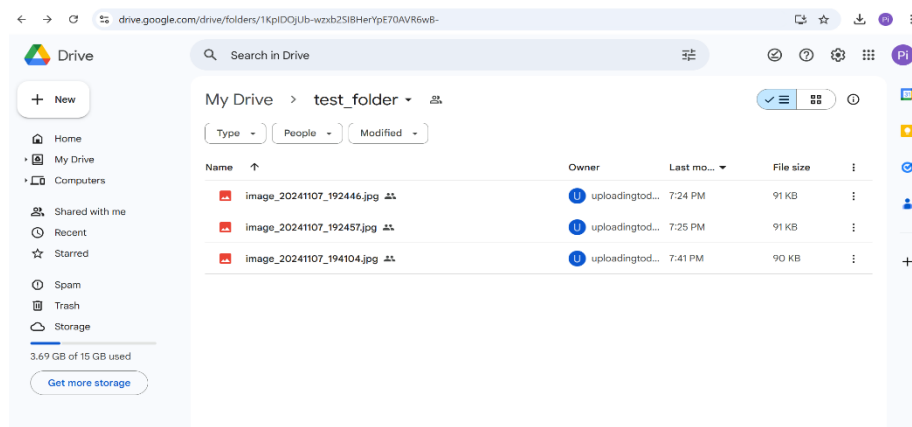


Fig 5.9 Images uploaded in drive

CHAPTER 6

RESULTS



Animal image captured by USB web cam

**Animal Detected: Cheetah
at 2024-11-09 12:25:09
follow the link to see the
image
[https://drive.google.com
/drive/folders/1KpID0jUb
-wzxb2SIBHerYpE70AVR6w
B-?usp=drive link](https://drive.google.com/drive/folders/1KpID0jUb-wzxb2SIBHerYpE70AVR6wB-?usp=drive_link)**

Alert notification sent through SMS

You can follow the given link to see the image that captured by the model and uploaded in drive through the following drive link.

[https://drive.google.com/drive/folders/1KpID0jUb-wzxb2SIBHerYpE70AVR6wB-
?usp=drive link.](https://drive.google.com/drive/folders/1KpID0jUb-wzxb2SIBHerYpE70AVR6wB-?usp=drive_link)

In conclusion, our project, the Animal Detection and Alert System, captures images of detected animals and sends SMS alerts to registered mobile numbers.

CHAPTER 7

ADVANTAGES OF THIS MODEL

- **Enhance Wildlife Protection**

The system helps in monitoring the presence of animals, especially harmful or endangered species, allowing authorities to respond swiftly and reduce human-wildlife conflicts.

- **Real-Time Alerts and Notifications**

By automatically sending SMS alerts when specific animals are detected, the project increases safety for both humans and animals, enabling rapid responses to potential threats.

- **Automated Data Collection**

The system captures images and data on animal activity without the need for constant human supervision, providing valuable insights into animal behavior and movements for researchers and conservationists.

- **Environmentally Friendly Monitoring**

With its minimal power requirements and remote capabilities, this system has a low environmental impact, making it suitable for sensitive forest and nature reserve environments.

- **Remote Accessibility**

Using SSH protocol enables secure, remote control of the module from any location, providing real-time access for monitoring and adjustments without the need for physical presence, which is invaluable for projects in hard-to-reach areas.

- **Improved Security**

SSH is a secure protocol that encrypts data during transmission, ensuring the safety of sensitive data. This is essential when dealing with wildlife monitoring data and maintaining the integrity of the system in remote locations.

CHAPTER 8

FUTURE WORKS AND IMPROVEMENTS

- **Integration with GPS for Location Tracking**

Adding a GPS module to the Raspberry Pi setup would enable the system to record the precise location of each detection. This could help in mapping animal movements, identifying migration patterns, and locating animals in real time for better response and intervention.

- **Addition of Thermal or Night Vision Cameras**

Implementing thermal or night vision cameras would improve animal detection in low-light or night-time conditions, making the system effective 24/7. This would be particularly useful for monitoring nocturnal animals or in heavily forested areas where visibility is limited.

- **Multi-Sensor and PIR sensor Integration for Enhanced Detection**

Adding additional sensors, such as PIR sensors, motion sensors or acoustic sensors, could enhance detection accuracy. This multi-sensor approach could help reduce false positives and improve the reliability of the system by cross-verifying detections from multiple sources.



PIR sensor

- **Expansion of the Dataset for Enhanced Model Accuracy**

Future improvements could involve collecting more diverse and comprehensive datasets of animals. This would improve the CNN model's accuracy in identifying various species and enable it to classify a broader range of animals, including regional or rare species.

CONCLUSION

This project presents an innovative method for monitoring and safeguarding wildlife through the use of affordable technology like Raspberry Pi and machine learning. By implementing a type of artificial intelligence known as a Convolutional Neural Network (CNN), the system can effectively recognize and categorize various animals within a forest environment. It combines sensors and a USB webcam to facilitate real-time tracking of animals, and it can also notify authorities if any potential dangers arise.

The primary aim of this initiative is to enhance the protection of wildlife. By identifying potentially harmful animals, the system plays a crucial role in minimizing conflicts between humans and wildlife, thereby creating a safer environment for both. Additionally, it supports environmental research by collecting vital information about endangered species, which can help scientists and conservationists better understand animal behaviours and movement patterns.

What makes this project particularly noteworthy is its demonstration of how low-cost technology can be leveraged to address significant challenges in wildlife conservation. By making such tools available, more individuals and organizations can participate in efforts to protect our natural ecosystems.

Looking to the future, there are numerous opportunities for improvement. For example, increasing the accuracy of the animal recognition system could yield even more effective outcomes. Furthermore, incorporating additional data analysis features could provide researchers with richer insights into wildlife behavior and trends. Overall, this project has the potential to broaden its impact on wildlife monitoring and conservation, fostering a harmonious coexistence between humans and animals in our shared habitats.

REFERENCES

1. Prof. Shahshi Rekha, G Harish Babu A, Manjunath S Kautilya K Bhat. “Wildlife Monitoring Using Image Processing”. IJSTE - International Journal of Science Technology & Engineering | Volume 7 | Issue 01 | July 2020.
2. <https://www.kaggle.com/code/nimapourmoradi/animal-detection> for animal detection model.
3. <https://www.raspberrypi.com/documentation/computers/getting-started.html> raspberry pi setup
4. <https://www.raspberrypi.com/documentation/computers/configuration.html> configuring raspberry pi.
5. Seema Kumaria, Dr. Sumit Kumar Yadav. “Development of IoT Based Smart Animal Health Monitoring System using Raspberry Pi”. Special Issue based on proceedings of 4th International Conference on Cyber Security (ICCS) 2018.
6. <https://www.raspberrypi.com/documentation/computers/remote-access.html> for remote access of raspberry pi.
7. K. Shiva Prasad, A. Hemanth Reddy, M. Ravi Kumar Goud. “Motion Activated Wild Life Capturing”. International Journal of Engineering Science and Computing, April 2017.
8. youtube.